

KA2 Advanced Assignment Code:

```
#include <Servo.h>
#include <math.h>

const int POT_PIN = A0;
const int SERVO_PIN = 9;

// Servo calibration
const int PULSE_MIN = 600;
const int PULSE_MAX = 2400;
const int PULSE_MID = 1700;
const int SWEEP_AMPLITUDE = 700; // .1 degrees per count so 70 degree sweep

// Frequency sweep range
const float FREQ_MIN = 0.1;
const float FREQ_MAX = 4.0;
// .1 and 10 work best or 7 actually but showing jittering towards end
// Timing
unsigned long lastTime = 0;
float phase = 0.0;
float currentFrequency = 1.0;

// For amplitude tracking
int maxPulse = PULSE_MID;
int minPulse = PULSE_MID;
unsigned long lastResetTime = 0;
const int RESET_INTERVAL = 2000; // Reset min/max every 2 seconds

Servo myServo;

void setup() {
    Serial.begin(115200); // baud is higher than usual so theres
    // enough time for the position to change for the servo
    myServo.attach(SERVO_PIN, PULSE_MIN, PULSE_MAX);
}

void loop() {
    // Read frequency command
    int potValue = analogRead(POT_PIN);
    currentFrequency = map(potValue, 0, 1023, FREQ_MIN * 100, FREQ_MAX * 100) /
    100.0;

    // Update phase
```

```

unsigned long now = micros();
float dt = (now - lastTime) / 1000000.0;
lastTime = now;

phase += 2.0 * PI * currentFrequency * dt;
if (phase > 2.0 * PI) phase -= 2.0 * PI;

// Compute and send command
int commandPulse = PULSE_MID + SWEEP_AMPLITUDE * sin(phase);
commandPulse = constrain(commandPulse, PULSE_MIN, PULSE_MAX);
myServo.writeMicroseconds(commandPulse);

// Track actual achieved positions (for bandwidth detection)
if (commandPulse > maxPulse) maxPulse = commandPulse;
if (commandPulse < minPulse) minPulse = commandPulse;

// Report every 2 seconds
if (millis() - lastResetTime > RESET_INTERVAL) {
    int actualSweep = maxPulse - minPulse;
    int commandedSweep = 2 * SWEEP_AMPLITUDE;
    float trackingRatio = (float)actualSweep / commandedSweep;

    Serial.print(currentFrequency, 2);
    Serial.print(" Hz:");
    Serial.print(actualSweep);
    Serial.println("  $\mu$ s");

    // Reset tracking
    maxPulse = PULSE_MID;
    minPulse = PULSE_MID;
    lastResetTime = millis();
}

delay(10);
}

```